

22. Search a 2D Matrix

```
class Solution {
public:
    bool searchMatrix(vector<vector<int>>& matrix, int target) {
        if (matrix.empty() || matrix[0].empty()) return false;
        int m = matrix.size();
        int n = matrix[0].size();

        int lo = 0, hi = m * n - 1;
        while (lo <= hi) {
            int mid = lo + (hi - lo) / 2;
            int r = mid / n;
            int c = mid % n;
            int val = matrix[r][c];

            if (val == target) return true;
            else if (val < target) lo = mid + 1;
            else hi = mid - 1;
        }

        return false;
    }
};
```

The screenshot displays a coding platform interface with the following components:

- Problem List:** Shows the problem name and status (Accepted).
- Submissions:** Indicates 133 / 133 testcases passed, submitted by 'stockahuja' on Nov 14, 2025 02:34.
- Performance Metrics:**
 - Runtime:** 0 ms, Beats 100.00%.
 - Memory:** 13.34 MB, Beats 42.03%.
- Code Editor:** Displays the C++ solution code for the 'Search a 2D Matrix' problem.
- Testcase Results:** Shows 'Accepted' status with a runtime of 0 ms. Two test cases are listed: 'Case 1' and 'Case 2'.
- Input/Output:** For Case 1, the input matrix is `[[1,3,5,7],[10,11,16,20],[23,30,34,60]]` and the target is `3`. The output is empty.